

Improving 3D-Editing Workflows via Acoustic Levitation

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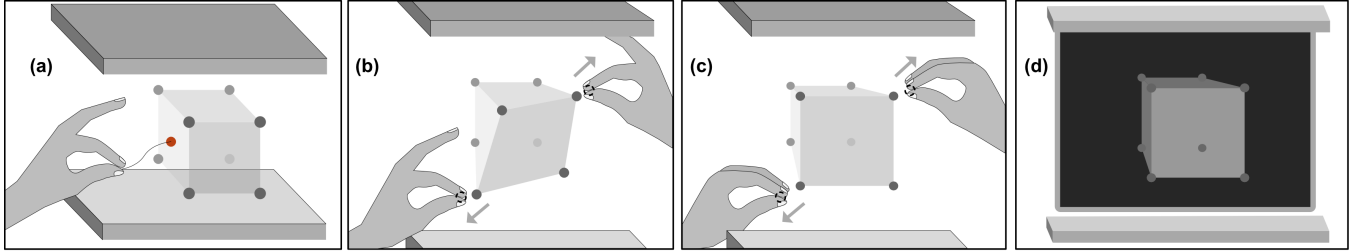


Figure 1: Interactive 3D display using acoustic levitation. (a) LeviProps provide custom handles for complex interactions like “Extrude”. (b) Pinch-grabbing two points move them and distort the surface, while (c) grabbing them scales the surface proportionally. A haptic sensation is rendered to indicate that beads are grabbed. High-resolution context could be displayed using AR glasses, a projector, or (d) an additional display.

ABSTRACT

We outline how to improve common 3D-editing workflows such as modeling or character animation by utilizing an acoustic levitation kit as an interactive 3D display. Our proposed system allows users to directly interact with models in 3D space and perform multi-point gestures to manipulate them. Editing of complex 3D objects can be enabled by combining the 3D display with an LCD, projector or HMD to display additional context.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

acoustic levitation, 3D editing, animation, multi-point interaction, mid-air interaction, tactile feedback

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1 INTRODUCTION

3D editors for modeling and animation, such as Blender [9] or Maya, [1] have to rely on input and output devices made for 2D content, requiring users to navigate 3D space. They need to find a mapping

between their 3D model space and the 2D display space. This mapping complicates interaction as it requires the user to learn how to navigate the scene and move and manipulate objects with 6 degrees of freedom within their 3D space. Existing solutions use AR or VR systems that do not require this mapping, e.g. Google Blocks [10]. However, they do not allow for spontaneous collaboration as an application displayed on a 2D display would do. Furthermore, those systems often lack sufficient haptic feedback. We propose using an acoustic levitation kit to physicalize control points for manipulating 3D objects. We identify the following possible contributions with our planned prototype:

- Explore various types of direct multi-point interaction using acoustic levitation, including gesture input and custom handles
- Combine the acoustic levitation kit with other input/output hardware to overcome technical limitations such as limited spatial resolution, therefore enabling a broader spectrum of interaction with devices based on acoustic levitation

2 RELATED WORK

Direct Manipulation & Touch Interaction

Shneiderman describes principles for direct manipulation interfaces [24, 25] that must be considered, especially when designing interactions constrained by technical limitations. Insights about precise touch interaction by Benko et al. [7] could be partially transferrable to interaction with a 3D display. ZeroN demonstrates tangible mid-air single-point interaction [15].

Additional Context through Secondary Display

Baudisch et al. [4–6] demonstrated with “Focus plus context screens” how an additional spatial context can be displayed in lower resolution to the user in their peripheral view. We plan to utilize a

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secondary display, projector or HMD to provide additional high-resolution context to the 3D editable but low-resolution model.

3D Modeling and Animation

When using a 2D screen, mapping must be applied between the screen and world space. The Cage [3] aids users in navigating 3D space. Paper3D [20] introduces a 3D modeling paradigm for multitouch interfaces. 3D puppetry [13] uses tangibles to animate the movement of rigid 3D objects in real-time. TinkerCAD [2] uses 3D primitives as building blocks and boolean operations to enable beginners to perform otherwise complex modeling tasks.

Volumetric Displays

Grossman et. al. found that depth perception on volumetric displays is significantly improved compared to stereographic displays or projection onto 2D displays [11]. Grossman et. al. demonstrate multi-point interaction with volumetric displays for 3D editing workflows by introducing touch- and gesture-based pointing techniques [12].

Acoustic Levitation and Acoustic Pressure Haptics

A volumetric display using acoustic levitation is demonstrated by Hirayama et al. [22]. Ultrahaptics [8] and HaptoMime [18] utilize acoustic radiation pressure to render haptic feedback mid-air. Hirayama et al. [21] developed a technique to compensate for distortions in the sound field produced by physical objects. LeviProps [19] are custom tangible structures we could use as visual indicators, tool selection, or menu items in our editor. Marzo et al. [16] show direct tangible interaction with floating particles.

3 OUR APPLICATION

Our proposed system consists of an acoustic levitation kit for multi-point interaction and a secondary display, projector, or HMD for displaying the model in high-resolution and providing additional spatial context.

3D Editor

We imagine a 3D editor where levitated beads act as control points for a model's geometry. Additional detail can be displayed using a secondary display for context, as described in the next section. Users can grab those control points (as described in the next section) and perform single-point tasks such as moving a bead or bi-manually perform multi-point tasks such as scaling or rotating the model. To differentiate between otherwise ambiguous actions such as “scale the whole model” in contrast to “scale that edge” (both performed by moving two beads simultaneously) we could introduce a small set of gestures (e.g. grab with thumb + index finger affects edge while grab with all fingers affects the whole model) and/or we could provide the user with selectable tools (“Move”, “Scale”, “Extrude”, ...) using LeviProps [19]. LeviProps could also be used to display custom action handles. For example, selecting “extrude” could spawn additional handles, represented by beads with an attached string to afford to pull. When pulling the string, the marked surface is extruded, as shown in figure 2. The movement

of the bead could be tracked by the camera since the user is not likely to occlude it while pulling the string. It is possible to display multiple handles or different kinds of handles simultaneously. Handles could enter the scene on tool selection or be brought in by the user. Also, handles could be used to output editor information (e.g. scaling factor) by projecting onto the attached fabric [19, 22].

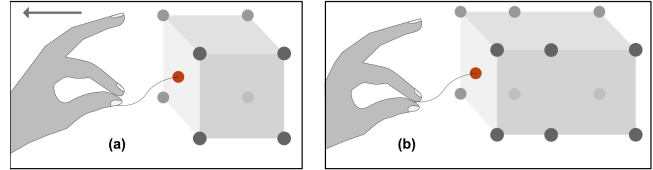


Figure 2: An extrusion handle (floating bead with attached string) is presented to the user to perform an extrude operation. (a) As the user pulls the handle, the manipulated surface is extruded, (b) adding additional geometry (floating beads are added). A haptic sensation is rendered on pulling.

The proposed editor could offer 3D primitives with triangular surfaces such as tetrahedrons and octahedrons (like TrussFab [14]). This allows adding additional geometry to the model while introducing very few new beads. As the number of beads that can be moved individually at the same time is limited, an editor based on tetrahedrons is suitable as it requires fewer beads for displaying comparable models than a boxel-based approach would need as shown in figure 3.

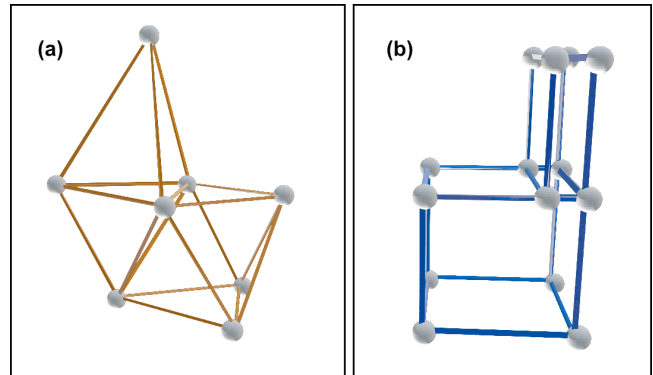


Figure 3: Wireframes of a simple chair model (a) made out of tetrahedrons and octahedrons can be displayed with only 8 vertex beads while (b) a comparable model made out of boxels would need 14 beads to be displayed.

Skeletal Character Animation

Likewise, our proposed interactive 3D display can be used for animating rigged 3D characters. Movements of character skeleton points can be recorded to enable character animation in real-time as opposed to keyframe-based animation common in other animation programs like [1] or [9]. The character itself is displayed in context as described below.

4 TECHNICAL CHALLENGES

Grab & Move Interaction

Ideally, interactions are designed in a way that allows (almost) directly grabbing beads while minimizing distortion of the acoustic field. Inserted hands can be detected via camera [23], i.e. a Leap Motion Controller (uses an infrared camera) [26]. Haptic feedback is rendered via acoustic radiation pressure [8, 18] to indicate that a bead was grabbed. As directly grabbing the bead might not be possible due to technical limitations, an alternative would be to select the bead by performing a grab/pinch gesture near the bead's position and render haptic feedback [16] as shown in figure 4. Alternatively, interaction with floating particles could be performed with acoustically-transparent tools or via pointing techniques [12].

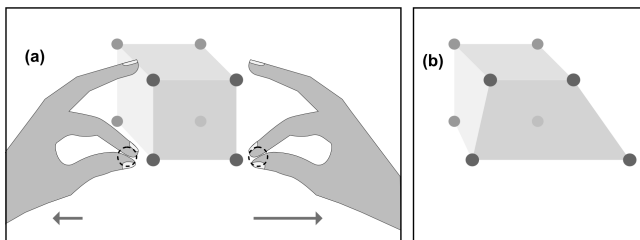


Figure 4: (a) The user performs a pinch close to the beads and pulls them apart. A haptic sensation is rendered to indicate that beads are grabbed. (b) Manipulated points are moved.

High resolution

A limited amount of additional geometry (e.g. edges) could be rendered by illuminating moving beads [22] or by oscillating a vertex bead in the direction of the vertex' adjacent edges as shown in figure. To allow editing of high-resolution 3D content, we could combine an acoustic levitation device with traditional monitors or projectors. This would allow users to interact with a resolution-reduced 3D representation while seeing a high-resolution representation displayed on an LCD. The acoustic levitation display would then allow for direct manipulation of an object's defining geometry, while the object can be rendered in full detail (textures and so on) and shown on an LCD. Alternatively, a combination with other methods of holographic imagery (i.e. AR glasses like HoloLens [17]) is possible to overlay the manipulable low-res version of the model with its full resolution version. Furthermore, both approaches would make it possible to zoom-in on more complex models while still being able to see the full model in context, similarly to "Focus plus context screens" [4–6].

5 CONCLUSION

In this proposal, we outlined an interactive system that utilizes acoustic levitation to facilitate 3D editing workflows.

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